



The influence of age, schooling, literacy, and socioeconomic status on serial-order memory

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Abstract We aimed at investigating whether formal schooling and literacy favor progress in verbal serial-order short-term memory (STM). In Experiment 1, we presented children varying on age, school level, and socioeconomic background with a serial-order reconstruction task and observed differences related to both age and school level. Two subsequent experiments aimed at separating schooling- and literacy-related effects from age-related ones. In Experiment 2 we compared, on the one hand, performance of children of similar age but different school levels and, on the other hand, performance of children of same school level but different ages. We observed a schooling effect but no age effect on serial-order reconstruction: the youngest first-graders outperformed age-matched kindergartners but performed similarly as older first-graders of the same

literacy level. Furthermore, children's literacy abilities were strongly correlated to their serial-order reconstruction performance, even after controlling for the effects of non-verbal reasoning and vocabulary. In Experiment 3 we examined low socio-economic background adults presenting varying (correlated) levels of schooling and literacy: some had attended school in childhood for only a few years and were either illiterate or very poor readers, whereas others had attended school for at least 12 years. In addition to serial-order memory, item STM was assessed by a delayed single nonword repetition task. The groups differed on both STM tasks, which both correlated with their literacy abilities. Thus, literacy and schooling do not impact only order memory, but more generally verbal STM. Moreover, comparison between the adults' and children's data strongly suggests that it is schooling and/or literacy rather than age per se that matters for serial-order memory.

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Introduction

The ability to temporarily retain ordered verbal sequences is usually related, not only to oral vocabulary acquisition (e.g., Baddeley et al. 1998; Leclercq

and Majerus 2010; Majerus and Boukebza 2013; Majerus et al. 2006), but also to reading and spelling acquisition (e.g., Binamé and Poncelet 2016; Martinez-Perez et al. 2012; Nithart et al. 2011; Schraeyen et al. 2017). According to an influential view, temporary memory acts as a bottleneck for learning, with weak memory capacities leading to learning difficulties, including in literacy acquisition (e.g., Arrington et al. 2014; Gathercole et al. 2004, 2006; Nevo and Breznitz 2011; Swanson and Jerman 2007).

Without denying this view, Demoulin and Kolinsky (2016) argued for an additional, reverse, relationship, with school activities bolstering short-term memory (STM). In particular, Demoulin and Kolinsky suggested that literacy acquisition impacts the recall of serial order: when decoding, the beginning reader has to internally produce the letter “sounds” and serially rehearse these to form a word, which may bolster serial-order processes. Under this view, decoding activities, due to their sequential nature, provide opportunities for training memory skills and strategies used to refresh the decaying phonological representations, such as cumulative rehearsal, namely the process of repeating to oneself (either overtly or covertly) the material to be remembered in serial order (see also e.g., Ellis 1990; Wagner 1974). Subvocal articulatory rehearsal would be particularly important for the encoding and maintenance of serial order information, as reflected, for instance, by the greater detrimental effect of irrelevant speech and articulatory suppression on the recall of serial order information than on the recall of item information (Henson et al. 2003). This account is consistent with the fact that order memory seems to have a specific impact on the acquisition of the sublexical reading and spelling routes. Indeed, in a 3-years longitudinal study on typically developing children, Binamé and Poncelet (2016) showed that serial-order memory was a robust predictor of nonword reading and spelling abilities in first and second grades, while being unrelated to the abilities to read and spell words.

This impact of literacy acquisition on the use of strategies that help maintaining a sequence of items in memory (such as cumulative rehearsal) might add to other literacy effects on verbal STM. For instance, literacy acquisition might improve the processing and the temporary encoding of spoken language through phoneme awareness, the ability to attend to or

manipulate the individual phonemes within spoken sequences. Although some specific activities at home or at kindergarten may favor the development of phoneme awareness (for a review, see Sénéchal 2015), this ability generally emerges when people begin reading lessons in an alphabetic code (e.g., Liberman et al. 1974; Lukatela et al. 1995; Morais et al. 1979; Read et al. 1986). In other words, learning to read boosts the acquisition of a system of explicit representations of speech, which could, in turn, increase the quality of phonological coding in verbal memory tasks. Consistent with this view, Melby-Lervag and Hulme (2010) showed that training 7-year-old typically developing children to manipulate phonemes (but not rhymes) in phonologically complex and unfamiliar words improves not only the phonemic manipulation of those words, but also their serial recall and, to a lesser extent, their free recall (see also Park et al. 2014, for consistent data on children with specific language impairment and reading difficulties). These findings suggest that phoneme awareness exerts qualitative changes on the underlying phonological representations, which in turn would lead to better memory encoding and/or retrieval.

Unfortunately, the impact of schooling and literacy acquisition on verbal STM, and more specifically on the ability to temporarily retain ordered verbal sequences, has seldom been investigated. Two reasons might explain the scarcity of the data. First, it is often difficult to distinguish the effects of age, socioeconomic status (SES), schooling and literacy on perceptual or cognitive processes (see discussion in e.g., Dehaene et al. 2015). Second, most STM tasks do not allow a precise assessment of serial-order recall (Hurlstone et al. 2014), as they mix several processes. For instance, both digit span and nonword repetition, the most frequently used tasks, require storing and retrieving two different types of information: the verbal items (the digits or phonemes/syllables) and their sequential order of presentation. Yet, a number of studies suggest that STM for item and order information reflect distinct cognitive processes. As a matter of fact, they are differentially impacted by psycholinguistic factors and interfering tasks. Indeed, semantic relatedness and word frequency, which reflect linguistic knowledge stored in long-term memory, influence recall of item identity to a stronger degree than recall of the

serial position of the items within a list (for a review, see Gorin et al. 2016). As already mentioned, other factors disrupt selectively STM performance for order relative to item information: when STM tasks are performed concurrently with interfering tasks (e.g., irrelevant speech, irrelevant tones, articulatory suppression, finger tapping), STM for serial order information is impacted to a greater extent than is maintenance of item information (e.g., Henson et al. 2003; for a review, see Gorin et al. 2016). Furthermore, STM for item and order information rely on different brain areas. More specifically, the to-be-remembered items would be stored via temporary activation of the language network in the temporal lobes, whereas a specialized system in the right fronto-parieto-cerebellar areas would deal with their order (Majerus et al. 2010).

To our knowledge, only one study examined specifically whether literacy acquisition impacts serial-order memory. Smalle et al. (2019) compared unschooled illiterate with schooled literate adults on a *serial-order reconstruction* task (Healy 1982; Healy et al. 1987; Nairne 1990) in which the items were simultaneously re-presented at recall in a random arrangement, so that only their presentation order needed to be reproduced. They observed better short-term serial recall performance in the literate compared with the illiterate people. Moreover, this effect was stronger in the verbal than in the visuospatial modality, suggesting that the improved capacity of the literate group is a consequence of literacy acquisition (although, potentially, it could also, or instead, be due to schooling).

However, the literate adults examined by Smalle et al. (2019) were Indian people who had acquired the Devanagari script, a Brahmic abugida (an alphasyllabary, akshara-based system) wherein consonants are placed linearly from left to right but vowels positioned non-linearly around them (above, below, or to either side of the consonants; see e.g., Joshi et al. 2017; Vaid and Gupta 2002). As a result, in some Devanagari words, the vowel precedes the consonant in writing but follows it in speech. In addition, reading in this script requires a quite large number of symbols to be acquired. Whether this and/or the (non-) serial complexity of the Devanagari script is at the source of the positive effect of literacy on serial-order STM reported by Smalle et al. is unknown. It would thus be useful to examine whether

a similar effect would hold true in people who have acquired or are in the process of acquiring an alphabetic script that uses only a limited number of linear symbols. In addition, as Smalle et al. contrasted two extreme groups, namely fully illiterate adults and schooled literates, they could not examine the correlations between memory performance and literacy level. In the present study, we aimed at examining such individual variability.

We therefore presented both either Brazilian Portuguese- or French-speaking children (Experiments 1 and 2) and adults (Experiment 3) with a serial-order reconstruction task inspired by the one designed by Majerus et al. (2006). This had been designed to maximize the short-term retention demands for order information while minimizing processing demands for item information. Indeed, item information is fully available at recall given that, after each sequence presentation, cards depicting the presented items are given to the participant who merely has to arrange them in their order of presentation. Moreover, the monosyllabic structure of the items minimizes phonological processing demands. These characteristics ensure the high sensitivity of the task to serial-order information and minimize the processing demands for item information. In addition, in Experiment 3 we also presented the participants with a control task that maximized processing requirements for item information while minimizing serial-order processing requirements. This allowed us to check whether literacy and/or schooling impact order memory in particular, or more generally verbal STM.

Experiment 1

In Experiment 1, we started by examining whether serial-order memory is associated to literacy abilities in Brazilian children of various school levels, from preliterate (kindergartners) to beginning readers (second-graders). We also examined children from various socioeconomic backgrounds. Indeed, several former studies have suggested that the quality of the childhood environment is crucial for STM development. For instance, Hackman et al. (2014) showed that lower parental education is associated with worse memory performance in children (see also e.g., Hackman and Farah 2009; Noble et al. 2007, for

consistent findings). Here, we examined both children attending private (paid) schools, which are usually frequented by families with middle to high SES, and children attending (free of charge) public schools, usually frequented by families with poorer SES (henceforth, *SES+* and *SES-*, respectively). We expected to observe both a general effect of type of school (associated to SES) and an effect of grade reflecting a strong correlation with literacy abilities. To examine the latter prediction, we presented the children with several ancillary tests: all were screened for letter and complex grapheme knowledge as well as for metaphonological abilities, including phoneme awareness. In addition, first- and second-graders were presented with five reading tests aimed at evaluating decoding abilities, reading fluency, and reading comprehension.

Method

Participants

A total of 120 children (57 boys and 63 girls) attending the last year of kindergarten, Grade 1 or Grade 2, participated in the study. All were Brazilian Portuguese natives. We discarded four children whose chronological age did not correspond to their school level (among whom one autistic child). Among the remaining 116 children (55 boys and 61 girls), none had any neurological or behavioral problem according to teachers' and parental reports. The children were attending one of three schools located in a small town in the South of Brazil (state of Rio Grande do Sul), one being a private school frequented by families with middle to high socioeconomic backgrounds, the other two being public schools frequented by families with poorer socioeconomic backgrounds. Consistently, familial income was higher in *SES+* than *SES-* schools¹: whereas the large majority (58.82%) of families of *SES-* schools reported an income of up to 3 minimum wage (MW),² this was the case of only 30.95% of the *SES+* families, who frequently

reported an income of 3–5 MW (28.57%) or even higher (5–10 MW: 23.81%; more than 10 MW: 16.67%). Although families of the *SES-* schools also reported an income of 3–5 MW quite frequently (35.29%), only 5.88% of them reported 5–10 MW and none reported an income of more than 10 MW. The parents' educational background (as they self-reported it in a questionnaire, see below) also differed as a function of SES. Level of education was coded as 0 if primary school was incomplete, as 1 if it was complete, as 2 if secondary school was complete, as 3 if higher education was in progress, as 4 if it was complete (undergraduate degree), and as 5 for a graduate degree. Although the mother's and father's levels of education were highly correlated, $\tau^{3(81)^4} = 0.50$, $p < 0.001$, we considered them separately, as the mother's level of education is usually reported as being more related to the children's development than the father's one, probably because she is usually the primary caregiver (e.g., Carneiro et al. 2013). Both the mothers' and the father's educational level was higher in the *SES+* than *SES-* schools: 3.514 (SD=1.484) vs. 2.102 (SD=1.342), and 3.236 (SD=1.427) vs. 1.596 (SD=1.014), respectively, both $ps < 0.001$ according to Mann–Whitney U tests.⁵ These average scores reflect the fact that about 10 times more mothers of the *SES+* schools' families than mothers of the *SES-* ones had a graduate degree ($\approx 41\%$ vs. 4% , respectively). Among mothers of the *SES-* schools, a majority ($\approx 47\%$) had finished secondary school, but $\approx 16\%$ had only finished primary school (0% in *SES+* schools) and 12% had even not finished it ($\approx 3\%$ in *SES+* schools). Not having finished primary school was also the case of $\approx 21\%$ of the fathers of the *SES-* schools' families, but of only 3% of the fathers of the *SES+* schools' families.

³ For all analyses, except for partial correlations we used the nonparametric Kendall rank correlation coefficient τ rather than the Pearson correlation coefficient r when the measure did not correspond to a ratio scale or when variances were unequal.

⁴ Many parents filled the questionnaire in an incomplete way, which explains the missing data.

⁵ Except for analyses of variance (ANOVAs) including several variables, for all group comparisons we used nonparametric tests when the measure did not correspond to a ratio scale (in which case we used the Mann–Whitney test) or when the two samples had significantly unequal variances and/or very unequal sample sizes (in which case we used the Welch test).

¹ As many parents filled the questionnaire in an incomplete way, these data were estimated on 93 families (42 from *SES+*, 51 from *SES-* schools).

² In August 2019, the MW was R\$ 998, which was equivalent to ≈ 247 US\$ (≈ 222 €).

Parental written consent was obtained for each child via letters explaining the purpose of the study and child oral consent was obtained on the first day of testing. In their respective schools, formal reading instruction started at first grade and used a mixed method combining phonics and whole language approaches.⁶ The kindergarten curriculum varied according to the schools and teachers, with some of them teaching the sounds of a few letters, but without formal reading instruction. The final samples included 41 kindergartners aged about 5 years and a half (21 SES+, 9 girls, aged 66.19 months on the average, SD=3.99, and 20 SES-, 9 girls, aged 68.45 months on the average, SD=3.53), 40 first-graders aged about 6 years and a half (20 SES+, 11 girls, aged 79.05 months on the average, SD=3.58, and 20 SES-, 10 girls, aged 83.15 months on the average, SD=3.15), and 35 second-graders aged about 7 years and a half (15 SES+, 11 girls, aged 90.53 months on the average, SD=3.31, and 20 SES-, 11 girls, aged 91.5 months on the average, SD=3.14).

All the children were presented with the following five ancillary tests, adapted to Brazilian Portuguese from those developed for a study commissioned by the National Reading Plan of the Portuguese Ministry of Education (Morais et al. 2010): letter knowledge (identification of 26 letters, each presented in both lower-case and upper-case, in two separate sub-tests), complex grapheme knowledge (38 items mixed into a single list: 31 lower-case digraphs, e.g., <lh>, and seven lower-case letters with diacritics, e.g., <ã>), as well as three metaphonological tests of varying

difficulty: phonemic sensitivity, initial syllable deletion, and initial phoneme deletion. In the phonemic sensitivity test, children were presented with six panels, each with six drawings of common objects. On each panel, they were asked to point the drawings corresponding to names of objects that started with a target phoneme. In the demo trial, for the phoneme /f/, they heard six words uttered by the experimenter: “fita”, “fada”, “ferro”, “fumo”, “fato”. Their attention was directed to the fact that all words started with the /f/ “sound”. They were next presented with six drawings and asked to point those corresponding to a name starting with /f/. On each panel there were from two to four images to be pointed (total: 18), half with a name starting with a simple (C) onset (e.g., “faca”), the others with a name starting with a complex (CC) onset (e.g., “flor”). The phoneme deletion task requires more than conscious recognition of phoneme identity, as it involves intentional manipulations of segments, hence leading to lower performance than phonemic sensitivity, even taking into account the theoretically different chance levels (e.g., Stanovich et al. 1984; Morais et al. 2010). Syllable deletion is far easier (e.g., Liberman et al. 1974). The syllable and phoneme deletion tests included 10 disyllabic items each (all CVCV in the phoneme deletion test), plus two training trials with corrective feedback. In both tests, children had to repeat part of a spoken pseudoword uttered by the experimenter after having deleted its initial part, either the first syllable (e.g., /kəbu/ - /bu/) or the first phoneme (e.g., /təbu/ - /əbu/). All expected responses were also pseudowords.

A minority of first-graders and most second-graders were in addition presented with five reading tests: word reading (12 simple, 12 complex and 12 irregular words), pseudoword reading (8 simple and 8 complex pseudowords), word and pseudoword reading fluency (two separate lists of 300 items each, with items of 1–3 syllables of various levels of complexity, each to be read as rapidly as possible in 1 min), and reading comprehension. All the reading tests were adapted from Morais et al. (2010), except for the reading comprehension test, which was a Portuguese-validated version (Sucena and Castro 2010)⁷ of the *Lecture 3* test originally designed in French by Lobrot (1973). This consists of 40 items (4 being

⁶ As recently discussed by Castles et al. (2018, p. 5), “over many years, the pendulum has swung between arguments favoring a *phonics* approach, in which the sounds that letters make are taught explicitly (e.g., Chall 1967; Flesch 1955), and a *whole-language* approach, which emphasizes the child’s discovery of meaning through experiences in a literacy-rich environment (e.g., Goodman 1967; Smith 1971)”. Although teaching phonics is crucial because it allows beginning readers to translate the spellings of most words they encounter into sound, and hence to become rapidly autonomous readers (e.g., National Reading Panel 2000), most alphabetic codes present at least some degree of spelling-to-sound irregularity. Hence teachers often choose to teach “sight words” along with phonics, i.e., to teach the pronunciations of a small number of words directly, particularly those frequently found in texts.

⁷ At the time we planned the research, the Brazilian version of the test was not available (de Araújo Vilhena et al. 2016).

training items with feedback) of increasing difficulty, formed by single incomplete sentences, each followed by a choice among five words as alternative completions: one that correctly completes the sentence, and four distractors that share phonological, orthographic or semantic similarity with the target word, or none. The child's task is to silently read these incomplete sentences and select the word that is meaningful within the sentence within a 5-min time limit for the 36 test sentences. Thus, this test assesses both reading speed and reading comprehension.

Parents were administered a questionnaire about their educational background, profession, average income, type of reading material they have at home, how often and who reads to or with the child, if the child could already read (in the case, when/how s/he has learned), reading preferences, kindergarten and/or school entry age, sleeping habits, hobbies, favorite games, as well as about their child's language(s), history of neurological disorders, neurodevelopmental delay, sensory impairment, speech or language impairment or learning difficulties.

Experimental materials

The task was an adapted version of the serial-order reconstruction task designed by Majerus et al. (2006). As we wanted to use only monosyllabic highly familiar and imageable words, in Brazilian Portuguese it was not possible to include only animal names and hence to present the task as an animal race, as Majerus et al. did. We rather selected seven highly familiar imageable words belonging to several semantic categories: flor, sol, nó, pé, cão, rã, boi, meaning “flower”, “sun”, “knot”, “foot”, “dog”, “frog” and “ox”, respectively. Their mean lexical frequency was very high (870.57) according to a Brazilian lexical database (*LexPorBR*, Estivalet and Meunier 2015).

These seven stimuli were used to form lists with lengths ranging from two to seven items. There were four trials for each list length. For each list (i.e., trial), the items were randomly selected from a pool of seven, and no item could occur twice in the same list. The stimuli were presented by increasing list length, beginning with list length 2.

The procedure was as follows. The stimulus list was pronounced by the experimenter, who was a trained Brazilian native speaker. The mean duration

of individual words was 495 ms (range 625–331 ms); they were presented in the list with an interstimulus interval of about 500 ms. After the auditory presentation of the list, the child was given 6 × 6 cm cards. On each card, one of the items whose oral names the child had heard was depicted. Only the cards for the items actually presented were given to the child. Thus, for list length 2, the child received two cards; for list length 3, the child received three cards, and so forth for subsequent list lengths. The child had to arrange the cards (given in random order) following the order of presentation of the auditory sequence by putting them on a 50 × 66-cm sheet on which a staircase-like figure with seven steps was depicted. The child had to put the first item of the list at the highest step, the second item at the second-highest step, and so forth for the subsequent items. However, the order of reconstruction was free: the child could begin order reconstruction at any serial position. For list lengths below seven items, the unneeded steps were covered by a blank sheet. After the child's response (which was written down by the experimenter), the experimenter removed the cards and presented the next list.

The child was told the following story for task description: *Every year, in a crazy school, on children's day, children get crazy gifts, winning from two to seven gifts, depending on their age. Look at the possible gifts.* Then the experimenter showed the child each of the cards corresponding to each of the gifts, and then began the test by an example: *I will tell you what the children will win and in what order, from the first to the last gift. After listening, you will place the gift cards on this staircase, placing the first gift on the highest step, then the second on the second step, and this for all the cards.*

Note that the researcher warned the child when the number of “gifts” increased (e.g., *Now there will be three gifts for 3-year-olds*). The test was stopped if the participant failed to reproduce three out of the four items used for each list length.

Procedure

This study was approved by the Ethic Committee of Santa Cruz do Sul University (registration no. 34703314.9.0000.53430). The children were tested in a quiet room of their respective schools. The tests analyzed in the present paper are part of a larger

study including 14 tasks, presented in five individual sessions lasting approximately 30 min each, that were administered on different days over a few weeks. The ancillary tasks were presented at session 1, from mid-May to the beginning of July,⁸ in the following order: laterality preference (Agostini and Dellatolas 1988; Oldfield 1971); lower-case and then upper case letter knowledge; complex grapheme knowledge; word and pseudoword reading; reading comprehension; word and pseudoword reading fluency; phonemic sensitivity; syllabic and then phonemic deletion. The serial-order memory test was presented a session 4, from the beginning of August to mid-September, due to winter break in the second fortnight of July. At the end of the last session, the children were offered a candy as reward.

Results and discussion

Except if mentioned otherwise, all the computations and analyses presented in the paper were performed using the open-source graphical statistical package *JASP* (JASP Team 2019, Version 0.9.1, <https://jasp-stats.org/>).

Ancillary tests

Table 1 presents average performance on ancillary tests. For letter knowledge, complex grapheme knowledge, phonemic sensitivity, syllable deletion and phoneme deletion, separate analyses of variance (*ANOVAs*) taking school grade and type of school (SES+, SES−) as factors showed a significant main effect of grade, $F(2,110)=74.839, 28.874, 57.42, 76.758, \text{ and } 32.102$, respectively, all $ps < 0.001$ ⁹ ($\eta_p^2 = 0.576, 0.429, 0.511, 0.583, \text{ and } 0.369$, respectively). Post-hoc comparisons (with Bonferroni–Holm correction to counteract the problem of multiple comparisons) showed that all grades differed from each other on all these tests, all $ps < 0.01$. There were neither significant effects of type of school nor significant interactions between grade and type of school, all $ps \geq 0.20$. Second-graders from SES+ and SES− schools also performed similarly in word reading fluency, pseudoword reading, word reading

fluency, pseudoword reading fluency, and reading comprehension, all $ps > 0.25$. Yet, there was a significant correlation between the mothers' education level and pseudoword reading fluency, $\tau(33)=0.34, p < 0.025$.

Serial-order reconstruction

Average performance, separately by grade and type of school, is presented in Table 1. All analyses were performed on the total number of sequences accurately reconstructed. Yet for sake of comparison with other studies, and as this score is easier to interpret, in Table 1 we also present the average span score, calculated as the longest sequence on which the child had 100% correct, plus the proportion of trials succeeded on the next sequence length.¹⁰

An ANOVA ran on the total number of sequences accurately reconstructed took school grade and type of school as factors. It showed a significant main effect of grade, $F(2,110)=15.685, p < 0.001, \eta_p^2 = 0.222$, with, as expected, kindergartners displaying lower scores than first- and second-graders, $t(79)=4.285, d=1.018$ and $t(74)=5.208, d=1.243$, respectively, both $ps < 0.001$ (Bonferroni–Holm-corrected). Grade 1 and Grade 2 children did not differ from each other, $t(73)=1.086, p=0.28$ (Bonferroni–Holm-corrected), $d=-0.219$. The main effect of school type tends towards significance, $F(1,110)=3.483, p=0.065, \eta_p^2 = 0.031$, and there is a non-significant trend towards an interaction between school grade and type of school, $F(2,110)=2.39, p=0.096, \eta_p^2 = 0.042$. This trend reflects the fact that children from SES− schools presented a significantly lower performance than children from SES+ schools in kindergarten, $F(1,39)=3.925, p=0.05$, and that these SES groups also tended to differ in Grade 2, $F(1,33)=3.616, p=0.06$, but that there was no significant difference between the SES groups in Grade 1, $F < 1$. On the whole sample there were significant correlations between serial-order reconstruction performance and both the mothers' and the fathers' education level, $\tau(114)=0.18, \text{ and } =0.19$, respectively, both $ps < 0.05$.

Most importantly, on the whole sample of 116 children we observe strong and significant

⁸ The school year in Brazil starts in March.

⁹ Similar results were found with the Welch test when only school grade was taken into account.

¹⁰ This is equivalent to calculate the total number correct divided by 4 (as there were four sequence per length) plus 1 (as lists started with length 2).

Table 1 Experiment 1

School level	School type	Letter knowledge	Grapheme knowledge	Word reading	Pseudoword reading	Word reading fluency	1 min	Word reading fluency correct/1 min	Pseudoword reading fluency correct/1 min	Reading Comprehension Score	Phonemic sensitivity d'	Syllable deletion %	Phoneme deletion %	Serial order reconstruction
		%	%	%	%	%						%	%	Total Span correct
Kindergarten	SES–													
	Valid	20	7	0	0	0	0	0	0	0	20	20	20	20
	Missing	0	13	20	20	20	20	20	20	20	0	0	0	0
	Mean	35.77	3.76	–	–	–	–	–	–	–	0.72	2.5	0	6.3
	SD	[28.7]	[4.77]	–	–	–	–	–	–	–	[0.75]	[7.16]	[0]	2.58
3.75	Range	0–84.62	0–10.53	–	–	–	–	–	–	–	0.57–2.88	0–30	0–0	4–11
	SES+													2–
	Valid	21	7	0	0	0	0	0	0	0	21	21	21	21
	Missing	0	14	21	21	21	21	21	21	21	0	0	0	0
	Mean	33.97	0	–	–	–	–	–	–	–	0.66	2.38	0	8.67
4.25	SD	[27.16]	[0]	–	–	–	–	–	–	–	[0.87]	[10.91]	[0]	3.17
	Range	0–82.69	0–0	–	–	–	–	–	–	–	0.45–3.14	0–50	0–0	4–13
	SES–													2–
	Valid	20	19	7	7	7	7	7	7	7	20	20	20	20
	Missing	0	1	13	13	13	13	13	13	13	0	0	0	0
Grade 1	SES–													
	Valid	20	19	7	7	7	7	7	7	7	20	20	20	20
	Missing	0	1	13	13	13	13	13	13	13	0	0	0	0
	Mean	82.4	33.24	80.56	81.25	16.71	17.43	13.49	11.72	11.15	2.36	58	35	11.55
	SD	[15.49]	[40.67]	[14.87]	[20.41]	[9.79]	[8.324]	[11.72]	[4.72]	[1.18]	[1.15]	[35.63]	[4.72]	3.89
2–6.5	Range	44.23–100	0–94.74	50–	97.22	43.75–100	5–34	6–30	5.56–38.89	0.42–3.83	5.56–38.89	0–100	0–100	4–22
	SES+													
	Valid	20	15	7	7	2	2	2	2	2	20	20	20	20
	Missing	0	5	13	13	18	18	18	18	18	0	0	0	0
	Mean	70.58	59.12	43.65	33.04	20.5	18	6.94	1.96	1.01	2.17	49.5	15	10.7
Grade 1	SES+													
	Valid	20	15	7	7	2	2	2	2	2	20	20	20	20
	Missing	0	5	13	13	18	18	18	18	18	0	0	0	0
	Mean	27.02	[32.35]	[25.34]	[17.94]	[7.78]	[8.49]	[1.96]	[31.37]	[3.83]	[1.01]	[36.49]	[31.37]	[0.96]
	SD	[27.02]	[32.35]	[25.34]	[17.94]	[7.78]	[8.49]	[1.96]	[31.37]	[3.83]	[1.01]	[36.49]	[31.37]	[0.96]
Grade 2	SES–													
	Valid	20	15	7	7	2	2	2	2	2	20	20	20	20
	Missing	0	5	13	13	18	18	18	18	18	0	0	0	0
	Mean	70.58	59.12	43.65	33.04	20.5	18	6.94	1.96	1.01	2.17	49.5	15	10.7
	SD	[27.02]	[32.35]	[25.34]	[17.94]	[7.78]	[8.49]	[1.96]	[31.37]	[3.83]	[1.01]	[36.49]	[31.37]	[0.96]
Grade 2	SES+													
	Valid	20	15	7	7	2	2	2	2	2	20	20	20	20
	Missing	0	5	13	13	18	18	18	18	18	0	0	0	0
	Mean	70.58	59.12	43.65	33.04	20.5	18	6.94	1.96	1.01	2.17	49.5	15	10.7
	SD	[27.02]	[32.35]	[25.34]	[17.94]	[7.78]	[8.49]	[1.96]	[31.37]	[3.83]	[1.01]	[36.49]	[31.37]	[0.96]
Grade 2	SES–													
	Valid	20	15	7	7	2	2	2	2	2	20	20	20	20
	Missing	0	5	13	13	18	18	18	18	18	0	0	0	0
	Mean	70.58	59.12	43.65	33.04	20.5	18	6.94	1.96	1.01	2.17	49.5	15	10.7
	SD	[27.02]	[32.35]	[25.34]	[17.94]	[7.78]	[8.49]	[1.96]	[31.37]	[3.83]	[1.01]	[36.49]	[31.37]	[0.96]
Grade 2	SES+													
	Valid	20	15	7	7	2	2	2	2	2	20	20	20	20
	Missing	0	5	13	13	18	18	18	18	18	0	0	0	0
	Mean	70.58	59.12	43.65	33.04	20.5	18	6.94	1.96	1.01	2.17	49.5	15	10.7
	SD	[27.02]	[32.35]	[25.34]	[17.94]	[7.78]	[8.49]	[1.96]	[31.37]	[3.83]	[1.01]	[36.49]	[31.37]	[0.96]

Table 1 continued

School level	School type	Letter knowledge	Grapheme knowledge	Word reading	Pseudoword reading	Word reading fluency	Pseudoword reading fluency	Reading Comprehension	Phonemic sensitivity	Syllable deletion	Phoneme deletion	Serial order reconstruction	
		%	%	%	%	correct/1 min	correct/1 min	Score	<i>d'</i>	%	%	Total correct	Span
2–6.5	Valid	20	20	17	17	14	14	14	20	20	20	20	20
	Missing	0	0	3	3	6	6	6	0	0	0	0	0
	Mean	94.13	72.63	76.63	73.16	26.71	22.36	24.6	2.94	71.5	60	10.85	3.71
	SD	[4.21]	[31.43]	21.25]	[25.65]	[12.62]	[10.78]	[14.24]	[0.69]	[31]	[40.52]	[4.84]	[1.21]
	Range	82.69–100	2.63–100	36.11–	97.22	18.75–100	10–51	6–44	2.78–55.56	0.9–3.83	10–100	0–100	4–22
Grade 2	SES+												
	Valid	15	15	13	13	12	12	12	15	15	15	15	15
	Missing	0	0	2	2	3	3	3	0	0	0	0	0
	Mean	92.82	79.65	82.05	77.4	31.75	26.33	30.56	2.76	81.33	58.67	13.33	4.33
	SD	[12.56]	[28.84]	[17.66]	[21.73]	[19.49]	[9.3]	[13.35]	[0.97]	[23.86]	[40.33]	[3.94]	[0.99]
2–5.5	Range	51.92–100	10.53–100	36.11–	97.22	37.5–100	7–69	9–41	8.33–58.33	0.32–3.83	20–100	0–100	4–18

Average values on the ancillary tests and on the serial order reconstruction task (in bold). Standard deviation in brackets. The cells in italics signal when only a minority of children have been tested (see valid and missing values)

correlations between serial-order reconstruction performance and letter knowledge, grapheme knowledge, phonemic sensitivity, syllable deletion and phoneme deletion, $\tau(114)=0.40, 0.37, 0.37, 0.47$, and $=0.40$, respectively, all $ps < 0.001$. Performance on serial-order reconstruction was also strongly correlated with age, $\tau(114)=0.31, p < 0.001$, which itself strongly correlated with all reading scores, all $ps < 0.001$. Nevertheless, all the correlations between performance on serial-order reconstruction and the literacy scores remained significant after controlling for the effect of age, $r(114)=0.27, p < 0.01, =0.30, p < 0.01, =0.19, p < 0.05, =0.51, p < 0.001$, and $=0.38, p < 0.001$ for correlations between serial-order reconstruction and letter knowledge, grapheme knowledge, phonemic sensitivity, syllable deletion and phoneme deletion, respectively. This suggests that literacy and/or schooling, in addition to age, is correlated to memory for order performance. In agreement with this idea, the correlations between the literacy scores and performance on serial-order reconstruction were significant within the group of first graders [$\tau(38)=0.36, =0.31, =0.46, =0.48$, and $=0.36$ with letter knowledge, grapheme knowledge, phonemic sensitivity, syllable deletion and phoneme deletion, respectively, all $ps < 0.01$], despite the fact that their age varied only slightly and hence was not correlated with performance on serial-order reconstruction, $\tau(38)=0.07, p > 0.10$. This was also the case in the second-graders: their serial-order reconstruction performance was significantly correlated with letter knowledge, grapheme knowledge, syllable deletion and phoneme deletion, $\tau(33)=0.28, =0.30, =0.34$, and $=0.28$, respectively, all $ps < 0.05$, but not with age, $\tau(33)=0.16, p > 0.10$. In addition, in second-graders serial-order reconstruction performance was significantly correlated with both word and pseudoword reading fluency, $\tau(33)=0.30, p < 0.05$ and $=0.44, p < 0.005$, respectively. These correlations remained significant when controlling for phoneme awareness, $r(33)=0.42, p < 0.025$ and $=0.56, p < 0.001$, respectively.

These data suggest that schooling and/or literacy, in addition to age, are related to serial-order memory. Yet, this relation could reflect either the fact that memory capacities act as a bottleneck for academic achievement, among which reading acquisition (e.g., Arrington et al. 2014; Gathercole et al. 2004, 2006; Swanson and Jerman 2007), or the fact that school

activities themselves, and in particular literacy acquisition, bolster serial-order memory. In the next experiment, we examined the latter possibility.

Experiment 2

Using a serial-order reconstruction task similar to the one of Experiment 1, the present experiment aimed at examining more thoroughly the relation between children's serial-order memory and age-related vs. schooling- and literacy-related factors. With this purpose we recruited 6- and 7-years-old French-speaking children, using a *cutoff design*. This takes advantage of arbitrary school cutoff dates that determine when children of virtually the same age begin formal schooling (e.g., Bisanz et al. 1995; Burrage et al. 2008; Cahan and Cohen 1989; Cahan and Davis 1987; Christian et al. 2000; Morrison et al. 1995; Varnhagen et al. 1994). In Belgium, where this second experiment took place, primary education normally begins in September of the calendar year in which children reach the age of six. Thus, we can consider that the cutoff date for entering first grade is December 31: all children reaching 6-years-old age before or at this date could enter first grade, whereas those born just after this date usually attend their last year of pre-primary school.¹¹ This system creates a “natural experiment”, allowing distinguishing schooling effects from those related to age (e.g., Morrison et al. 1995). Indeed, participants born close to the cutoff date have different schooling experiences but are very close in chronological age. By contrast, children within the same class may largely differ in age, with, for instance, a child born in January and another, in December; whereas the former will be among the oldest of the class, the latter will be among the youngest.

In a previous longitudinal study with a cut-off design, Morrison et al. (1995) examined the effects of age and schooling on STM by comparing “young” first graders (henceforth, *Y1*) to “old” kindergartners

¹¹ Pre-primary education is very popular but not compulsory in Belgium, with more than 95% of children under the age of six attending it (Eurydice 2011). Children may enter pre-primary education at the age of two-and-a-half at the earliest and leave it in June of the year in which they reach the age of six.

(henceforth, *OK*). At the beginning and at the end of the school year, both groups were tested with a free recall task in which they had to verbally recall the names of pictures. Whereas there were no group differences at the beginning of the year, the Y1 outperformed the OK children in STM at the end of the school year. In addition, no memory enhancement was observed after 1 year at kindergarten. One year later, both groups were tested again (in Grade 1 for the original OK children and in Grade 2 for the original Y1 children): it was only after 1 year of formal education that children in the original OK group reached the level of the original Y1 group. These findings suggest that the growth of immediate memory skills depend more on formal schooling experience than on age maturation.

In the present experiment, we examined all children only once, at the end of the school year. As Morrison et al. (1995), we compared OK to Y1. In addition, we compared these Y1 to “older” first graders (henceforth, *O1*). We thus compared, on the one hand, children of similar ages but different schooling and reading experiences and, on the other hand, children of different ages but similar schooling and reading experiences. If formal education and associated experiences improved memory for order, we should observe better serial-order reconstruction performance in the Y1 group compared to the OK one. By contrast, if progress in memory for order were mainly due to chronological age, we should observe better serial-order reconstruction performance in O1 compared to Y1, but no difference between Y1 and OK. In line with the findings of Morrison et al., we predicted school-related effects on serial-order reconstruction performance, namely, better performance in Y1 compared to OK, without differences between Y1 and O1.

In order to examine more closely the relationship between literacy and serial-order memory performance, we evaluated the children’s literacy abilities through several reading and phonological awareness tests. We predicted literacy level to be correlated with serial-order reconstruction performance. We also checked for the relationships with children’s nonverbal reasoning abilities and vocabulary level.

Method

Participants

A total of 54 children (32 boys) participated in the study. They were all French native; none had any social, neurological, or behavioral problem according to teachers’ and parental reports. The children were selected from seven schools of the French community of Belgium located either in the urban area of Brussels or in the rural area of Wallonia. All children were born at the beginning or at the end of a civil year (within the 2 months and a half around December 31, the cut-off date). Parental written consent was obtained for each child via letters explaining the purpose of the study and child oral consent was obtained on the first day of testing. Parents were also administered a questionnaire ensuring that their child’s main language at home was French and that he/she had no history of neurological disorders, neurodevelopmental delay, sensory impairment, speech or language impairment or learning difficulties. Children who did not meet these criteria were excluded. Most children came from families with middle socioeconomic backgrounds. All the children were in the age-appropriate grade and all the first-grade children had attended kindergarten the previous years. In their respective schools, formal reading instruction started at first grade and used a mixed method combining phonics and whole language approaches. The kindergarten curriculum varied according to the schools and teachers, with some of them teaching the sounds of a few letters, but without formal reading instruction.

The children were classified into one of three groups according to their birth date and school level. Each group included 18 children. At the time of assessment (second term), the mean age of the OK was 75 months (74–76 months). On average, the Y1 were only 2 months older than the OK, with a mean age of 77 months (76–78 months). The O1 were about 10 months older than the Y1, with a mean age of 87 months (85–88 months), $p < 0.001$ according to a t -test.

The children were presented with six ancillary tests. Nonverbal reasoning was measured by Raven’s Colored Progressive Matrices (*RCPM*; Raven et al. 1998). Receptive vocabulary was assessed by the *Échelle de vocabulaire en images Peabody (EVIP*;

Dunn et al. 1993). Word and nonword reading as well as syllabic and phonemic awareness (measured through initial deletion tasks) were assessed with the *Batterie d'évaluation du langage écrit et de ses troubles* (BELEC; Mousty et al. 1994).

Experimental materials

A serial-order reconstruction task, *the animal race task* (Majerus et al. 2006), was used to measure STM for serial-order. The task was presented to the child as an animal race game and consisted of the auditory presentation of sequences of animal names (chat, chien, coq, lion, loup, ours, singe, meaning “cat”, “dog”, “rooster”, “lion”, “wolf”, “bear”, “monkey”) by increasing list length, from length 2 to 7. After each sequence, the child had to reconstruct the order of presentation of the names using cards depicting the animals. The seven stimuli were selected for their high lexical frequency and low age of acquisition (mean age of acquisition: 20 months; range 13–24 months) ensuring that they were highly familiar to the children. The experimenter presented the task as follows:

Every year, animals from all over the world gather to have a huge race. This year, seven animals are participating: a cat, a dog, a cock, a lion, a wolf, a bear, and a monkey [the experimenter shows the cards of the corresponding animals]. Several races take place. Sometimes only two animals are participating. Sometimes there are three, four, or five animals. At other times, there are big races with six or seven animals. Through the headphones, you will hear someone announce the animal's order of arrival at the finish line, from the first to the last animal. Immediately after, you have to put the pictures of the animals on the podium in their order of arrival. The animal arriving first has to be put on the highest step and the last one on the lowest step.

The stimuli were recorded by a trained French native female speaker, digitized and edited into individual files and presented binaurally via headphones connected to a laptop computer. Their mean duration was 549 ms (range 371–696 ms), with an interstimulus interval of 650 ms. The experimenter activated the presentation of each list (animal race), after which she gave the child the cards (in alphabetical order) that depicted the specific animals presented in the list. The child had to rearrange them

on a staircase with seven steps drawn on a sheet: he or she had to put the first item on the highest step, the second item on the second step, and so on. The experimenter wrote down the child's response and then removed the cards from the podium. Four different sequences were proposed for each list length but, as made by Majerus et al. (2006) in order to optimize the reliability of the estimate of a given child's performance level, the task was split into two equal parts, one part being presented in the first session, the other in a next session. The test was over if the participant failed to reproduce three successive sequences.

Procedure

This study was approved by the Ethic Committee of Université Libre de Bruxelles (registration no. 031/2012). To disentangle the effects of 1 year of schooling (which both the Y1 and the O1 experienced) from those of age-related development, the skills were assessed as close as possible to the end of the school year. The children were tested by the third author in a quiet room of their respective schools or of the university. Three individual sessions lasting approximately 35 min were administered on different days over few weeks. The first and second sessions started with the administration of the serial-order reconstruction task, followed by the receptive vocabulary and nonverbal reasoning tasks, respectively. The last session involved word and nonword reading, as well as syllabic and phonemic awareness tasks. At the end, the children were offered a book as reward.

Results and discussion

For all tests, we ran planned contrasts between, on the one hand, Y1 vs. OK, and on the other hand, Y1 vs. O1 (see average results and *p* levels in Table 2).

Ancillary tests

As can be seen in Table 2, The Y1 group did not significantly differ from the OK and O1 groups on nonverbal reasoning. Yet, Y1 performed better than OK and similarly to O1 on receptive vocabulary.

All the OK were pre-readers: none could read even any of the short words of the training list. The syllable and phoneme deletion abilities of the Y1

Table 2 Experiment 2

	Children			Planned contrasts <i>p</i> level	
	OK	Y1	O1	Y1 > OK	Y1 < O1
Non-verbal reasoning (raw scores)					
Mean	22.78	25.11	26.53	.11	.29
SD	[6.01]	[5.28]	[5.6]		
Range	14–33	19–34	16–34		
Receptive vocabulary (raw scores)					
Mean	82.06	90	92.67	.05	.26
SD	[16.14]	[12.51]	[11.88]		
Range	51–101	69–111	73–124		
Syllable deletion (%)					
Mean	65.11	97.94	90	.001	.93
SD	[35.78]	[4.81]	[21.96]		
Range	0–100	81–100	13–100		
Phoneme deletion (%)					
Mean	29.78	85.44	69.22	.001	.94
SD	[31.71]	[18.73]	[35.91]		
Range	0–88	46–100	0–100		
Word reading (%)					
Mean	0	69.67	63.28	— ^a	.78
SD	[0]	[14.85]	[30.88]		
Range	0–0	44–94	6–100		
Nonword reading (%)					
Mean	0	66.44	59.22	— ^a	.82
SD	[0]	[16.79]	[29.55]		
Range	0–0	33–88	8–92		
Overall literacy score (%)					
Mean	23.72	80.21	70.72	< .001	.91
SD	[15.46]	[11.14]	[26.61]		
Range	0–45.5	57–96	7.25–98.5		
Serial order reconstruction					
Total correct					
Mean	11	12.5	12.22	.04	.64
SD	[2.68]	[2.2]	[2.29]		
Range	8–19	8–16	8–15		
Span					
Mean	3.75	4.13	4.06		
SD	[0.67]	[0.55]	[0.57]		
Range	3–5.75	3–5	3–4.75		

Average values (in bold) on the ancillary tests and on the serial order reconstruction task. Standard deviation in brackets. Significant *p* values are in bold

^acf. 0 values in word and pseudoword reading in OK

were better than those of their age-matched peers in kindergarten and not poorer than to those of the O1. As a principal component analysis (PCA) ran on the four literacy-related tests (word and nonword reading, syllable and phoneme deletion) revealed only one underlying factor, Model $\chi^2(2)=85.218$, $p < 0.001$, we calculated a composite literacy score. Despite their age difference, Y1 were not poorer than O1 on this overall literacy score (see Table 2; similar results were obtained with Welch tests).

Serial-order reconstruction

School-level effects on memory performance in the serial reconstruction task were also assessed by running planned contrasts on the total number of sequences accurately reconstructed between, on the one hand, Y1 vs. OK, and on the other hand, Y1 vs. O1 (average results and p levels in Table 2). This revealed that the Y1 significantly outperformed the OK children on serial-order recall but did not differ from the O1, as also illustrated in Fig. 1.

On all children, performance on serial-order reconstruction was largely and positively associated with literacy abilities, non-verbal reasoning, and vocabulary, $\tau(52)=0.42$, $p < 0.001$, $=0.34$, $p < 0.001$, $=0.23$, $p < 0.025$, respectively, but not with age, $\tau(52)=0.12$, $p=0.26$. After controlling for the effects of non-verbal reasoning and vocabulary, partial correlations between serial-order memory and literacy abilities were still highly significant, $r(52)=0.38$ and $=0.40$, both $ps < 0.01$. Note that serial-order reconstruction performance remained highly significantly correlated with literacy abilities when considering only first-graders, $\tau(34)=0.46$, $p < 0.001$. In these children, serial-order reconstruction performance was significantly correlated with all literacy-related scores, $\tau(34)=0.38$, $=0.43$, $=0.45$ and $=0.36$ for word reading, pseudoword reading, phoneme deletion, and syllable deletion, respectively, all $ps \leq 0.01$. Note that pseudoword reading still tended to be correlated with serial-order reconstruction performance when controlling for the correlation with phoneme deletion, $r(34)=0.29$, $p < 0.10$.

In agreement with the results of Experiment 1, these data thus support the idea that the development of serial-order memory in children is more related to school activities, and in particular to literacy acquisition, than to age per se. They also strongly argue for

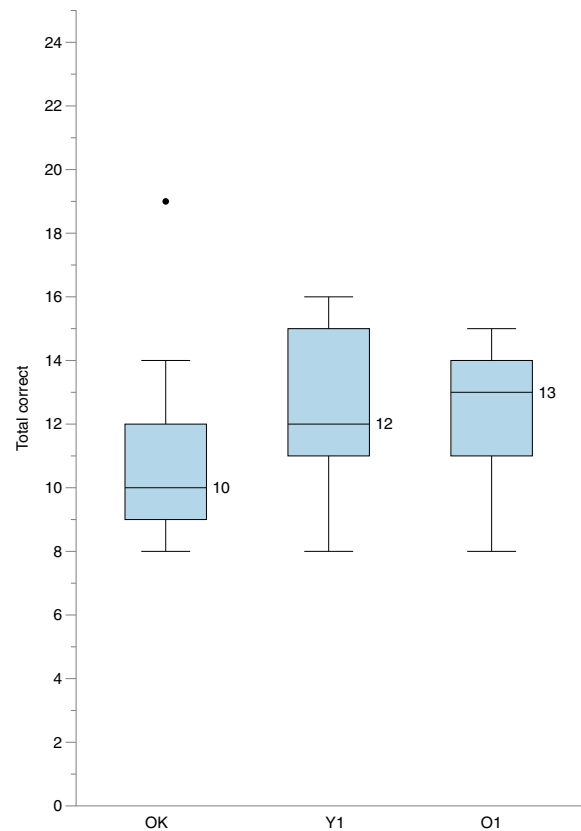


Fig. 1 Experiment 2: total correct score on serial order reconstruction in each group. The line on each boxplot denotes the median score (also presented numerically), the edges of the box denote the 25th and 75th percentile, and the whiskers extend the data points that lie within 1.5 times the interquartile range. The black dots represent outliers

the idea that schooling and literacy acquisition bolster serial-order memory. Otherwise, one can hardly understand why preliterate children display poorer serial-order memory performance than age-matched first-graders.

In the next experiment, we examined whether schooling level and literacy acquisition are also related to serial-order memory in adult participants.

Experiment 3

The data from Experiment 2 support the idea that school activities, and in particular literacy acquisition, benefit serial-order memory more than age per se. Yet they present several shortcomings. First, those results were obtained through the cutoff design,

which has been often criticized. Indeed, in addition to the logistic difficulty to find enough children in each group, this approach suffers from a serious selection bias, because the cutoff date is never strictly imposed, admission to school being sometimes delayed and sometimes accelerated. Moreover, those exceptions are not random: intellectually advanced children who are slightly younger than the official school age are often admitted, while children who are somewhat older than the cutoff point but judged insufficiently mature may be held out of formal schooling for an additional year and retained an extra year in kindergarten. Such a selection bias might have affected the results of Experiment 2, leading to an overestimation of the schooling effect (cf. e.g., Bentin et al. 1991; Cahan and Cohen 1989). Second, the age range contemplated so far was rather restricted.

For those reasons, in the present experiment we examined adults of low socio-economic background presenting varying levels of schooling and literacy. Some either had not attended school at all in childhood or had attended it in childhood for a few years (up to 5) and were either illiterate or very poor readers, whereas the others had attended school for 12–16 years. These two groups will be referred to as *–schooled adults* and *+schooled adults*, respectively. *All these participants were presented with 10 literacy-related tests in order to examine the relation between literacy level and serial-order memory performance. If schooling and literacy acquisition enhanced serial-order memory, we would expect +schooled adults to present better serial-order reconstruction performance than –schooled adults, and to observe a significant correlation between that performance and literacy level.*

In addition, in this experiment we aimed at checking whether literacy and/or schooling impact order memory in particular, or more generally verbal STM. We therefore used as control task a delayed single nonword repetition task that is supposed to maximize processing requirements for item information while minimizing serial-order processing requirements (e.g., Leclercq and Majerus 2010; Majerus et al. 2006). To maximize processing requirements for item information, the items were all different across the trials. By contrast, the need to retain order information was minimized for several reasons. First, only a single item had to be remembered, contrary to the serial-order reconstruction task

where several items and their arbitrary serial positions had to be remembered. Second, all the nonwords had exactly the same CVC monosyllabic structure, unlike traditional nonword repetition tasks typically using multisyllabic nonwords; hence it was mainly phoneme identity that had to be retained, as the only order errors that could occur were inversions between the first and last consonant. Third, sequential rehearsal of the to-be-stored information, assumed to be a major component of serial-order STM, was made difficult via articulatory suppression, namely the concurrent repetition of the syllable /blu/ during the retention delay.

Method

Participants

The samples included 35 adults of low SES presenting varying levels of schooling and literacy. Ten *–schooled* adults had either not attended school at all in childhood or attended it for a few years (up to 5; average: 2.2; SD=1.62); they were aged 59.1 years on the average, SD=8.07, 8 women (80%). The other participants were 25 *+schooled* adults who had attended school for 12–16 years (average: 13.16; SD=1.28); they were aged 41.32 years on the average, SD=7.5, 17 women (68%). The participants' income was associated with their number of school years, $\tau(33)=0.49$, $p < 0.001$: it was about 1.2 minimum salary (SD=0.95) in the *–schooled* adults and 3.48 (SD=0.51) minimum salary in the *+schooled* adults. The *–schooled* participants were also older than the *+schooled* ones, $t(33)=6.203$, $p < 0.001$, $d=2.321$.

Material and procedure

This study was approved by the Ethic Committee of Santa Cruz do Sul University (registration no. 34703314.9.0000.53430). Material and procedure were the same as in Experiment 1, except that all participants were presented with all the ancillary tests and also performed a single nonword repetition task, presented after the serial-order reconstruction one.

The single nonword repetition task consisted of 16 monosyllabic nonwords presented separately, and the items were new on any trial. All stimuli had a CVC syllabic structure and were legal with respect to

Portuguese phonotactic rules: they were constructed by changing the first consonant of an existing word (e.g., /sɔw/, meaning “sun”, became [fɔw]; /kajs/ or /pajs/, meaning “docks” or “parents”, became [najs]). On each trial, one nonword was presented and the participant was instructed to first repeat it immediately to confirm that he/she had correctly perceived the item and was able to reproduce it accurately; however, no corrective feedback was given. The participant was thereafter asked to continuously repeat the syllable /blu/ during 3 s. Then the experimenter instructed the participant to repeat again the stimulus. Performance was assessed on this delayed repetition only. As in former studies using this task (e.g., Leclercq and Majerus 2010), an item was scored as correct if production at recall was identical to what had been repeated immediately after presentation. This scoring method ensured that we assessed phonological short-term maintenance abilities without being biased by initial misperceptions.

Results and discussion

Ancillary tests

Table 3 presents average performance on ancillary tests. A PCA ran on the 10 literacy tests revealed only one underlying factor, Model $\chi^2(35)=707.624$, $p < 0.001$. We thus calculated an overall literacy score. On this score, +schooled adults clearly outperformed the –schooled ones, $t(9.697)=4.664$, $p < 0.001$ according to the Welch test¹².

Short-term memory tasks

As can be seen in Table 3, the +schooled adults significantly outperformed the –schooled adults on serial-order reconstruction, Mann–Whitney=227, $p < 0.001$.¹² In addition, there was a significant positive correlation between serial-order reconstruction performance and the overall literacy score, $\tau(33)=0.46$, $p < 0.001$. Yet, the –schooled participants might have been disadvantaged by the fact that they were older than the +schooled ones (see Participants

¹² This test was chosen because of the relatively small sample size of the +schooled participants ($N = 25$), in which there was a deviation from normality according to the Shapiro–Wilk test, $p < .05$.

Table 3 Experiment 3

Group	Letter knowledge	Grapheme knowledge	Word reading	Pseudoword reading	Word reading fluency	Pseudoword reading fluency	Reading comprehension	Phonemic sensitivity	Syllable deletion	Phoneme deletion	Overall literacy score	Serial order reconstruction		Item STM
	%	%	%	%	correct/1 min	correct/1 min	Score	d'	%	%		Total	Span	
-Schooled														
Mean	85.58	63.42	69.72	78.91 ^a	23.40	16	25.56	1.83	67	19	43.68	9.6	3.4	70.63
SD	[18.38]	[30.25]	[40.69]	[31.51]	[23.66]	[15.47]	[25.42]	[1.34]	[36.83]	[40.12]	[22.68]	[2.22]	[0.55]	[17.19]
Range	50-98.08	0-92.11	0-100	6.25-100	0-67	0-45	0-61.11	0.25-3.83	0-100	0-100	6.03-72.96	5-12	2.25-4	37.5-100
+Schooled														
Mean	99.54	87.58	99	96.75	83.44	49.56	83.44	3.14	88.4	86.8	77.77	14.96	4.74	95.50
SD	[1.01]	[10.81]	[1.58]	[5.14]	[17.61]	[15.65]	[12.31]	[0.66]	[21.92]	[23.58]	[7.02]	[3.93]	[0.98]	[6.38]
Range	96.15-100	60.53-100	94.44-100	81.25-100	56-123	25-87	55.56-100	1.78-3.83	20-100	0-100	59.95-90.12	9-22	3.25-6.5	75-100

Average values (in bold) on the ancillary tests and on the serial order reconstruction task. Standard deviation in brackets

^a Only 8 (out of 10) participants

section). As a matter of fact, there was a significant negative correlation between serial-order reconstruction performance and age, $\tau(33)=-0.37$, $p<0.005$. Nevertheless, the correlation between serial-order reconstruction performance and the average literacy score remained significant even after partialling out the effect of age, $r(33)=0.44$, $p<0.01$. Interestingly, serial-order reconstruction performance correlated significantly with both word and pseudoword reading, $\tau(33)=0.37$, $p<0.01$ and $=0.32$, $p<0.025$, respectively, as well as with both word and pseudoword reading fluency, $\tau(33)=0.47$ and $=0.46$, respectively, both $ps<0.001$. It also correlated significantly with reading comprehension, $\tau(33)=0.40$, $p<0.005$. Noticeably, these correlations cannot be accounted for only by the fact that serial-order reconstruction performance also correlated significantly with average phoneme awareness performance,¹³ the latter being also strongly correlated to average reading scores,¹⁴ $\tau(33)=0.45$, $p<0.001$. Indeed, serial-order reconstruction performance remained significantly correlated to the average reading score even after partialling out the effect of average phoneme awareness, $r(33)=0.42$, $p<0.025$ (more specifically, serial-order reconstruction performance remained significantly correlated to both word and pseudoword reading fluency, $r(33)=0.49$, $p<0.01$ and $=0.42$, $p<0.025$, respectively).

As can be seen in Table 3, the two groups also differed on item STM, with the +schooled adults significantly outperforming the -schooled adults, $t(10.01)=4.454$, $p=0.001$ according to the Welch test. Item STM performance also strongly correlated with the overall literacy score, $\tau(33)=0.41$, $p<0.005$. Although there was here also a significant negative correlation between item STM performance and age, $\tau(33)=-0.43$, $p<0.001$, the correlation between item STM performance and the average literacy score remained significant even after partialling out the effect of age, $r(33)=0.53$, $p=0.001$. The correlation between item STM performance and the average reading score also remained significant after partialling out the effect of average phoneme awareness, $r(33)=0.36$, $p<0.05$.

Thus, the effect of literacy on verbal memory was not restricted to serial-order memory. In addition, the item STM performance and serial-order memory performance were correlated to each other, $\tau(33)=0.36$, $p<0.01$, but this correlation did not remain significant after partialling out the effect of literacy, $r(33)=0.08$. Literacy performance thus seem to mediate the association between performance on the two STM tests.

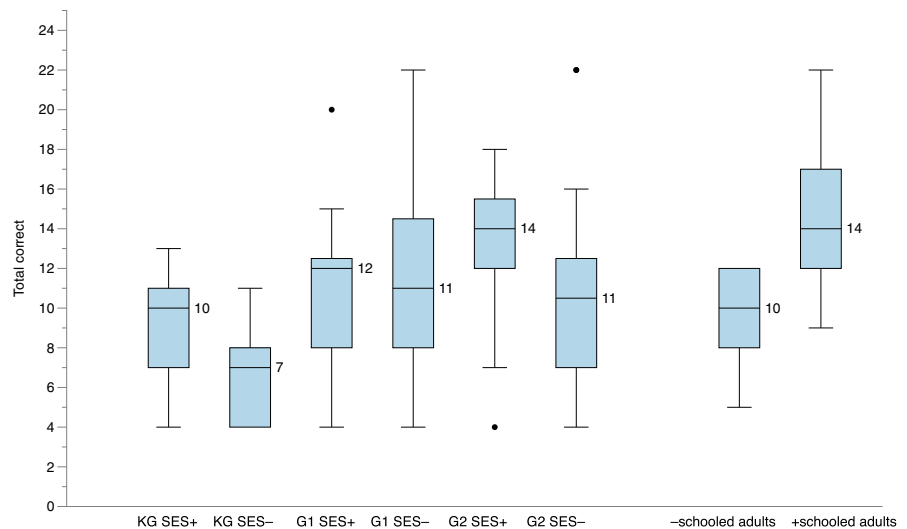
In the adults considered in the present study, average literacy performance was strongly associated to schooling level, $\tau(33)=0.55$, $p<0.001$. Thus, it is difficult to dissociate the effect of literacy per se from the effect of number of school years. Nonetheless, when considering the partial correlations between literacy and STM performance while controlling for the effect of number of school years, literacy tended to remain correlated with serial-order memory performance, $r(33)=0.28$, $p=0.10$, and remained significantly correlated to item STM performance, $r(33)=0.45$, $p<0.01$.

In any case, considering the whole lifespan between 5 years of age (Experiment 1) and adulthood (the present experiment), the data strongly suggest that it is literacy and/or literacy and schooling experience rather than age by itself that matters for serial-order reconstruction performance. Indeed, as illustrated in Fig. 2, the -schooled adult participants' performance seems closer to the performance of the children examined in Experiment 1 than to the one of +schooled adult participants. In fact, an omnibus one-way non-parametric ANOVA on ranks (Kruskal-Wallis test) ran on the serial-order reconstruction data of both Experiment 1 and the present experiment showed a highly significant effect of group, $H(4)=49.07$, $p<0.001$; using a parametric approximation, the group effect remains significant when taking age as a covariate, $F(4,145)=12.658$, $p<0.001$, $\eta_p^2=0.259$. Notably, the pattern of group differences was similar for serial-order reconstruction and for literacy performance. As a matter of fact, whereas the -schooled adults displayed slightly better serial-order reconstruction performance than the kindergartners, Mann-Whitney U tests $W=110.5$, $p<0.025$, they differed neither from the Grade 1 nor from the Grade 2 children, $W=251$, $p=0.22$, and $W=110.5$, $p=0.11$, respectively, and, similarly, the -schooled adults presented better average literacy scores only when compared to kindergartners, $W=16$, $p<0.001$, not

¹³ Computed on phoneme deletion and phoneme sensitivity.

¹⁴ Computed on word and pseudoword reading, word and pseudoword fluency, and reading comprehension.

Fig. 2 Total correct score on serial order reconstruction in each group of children (Experiment 1) and of adults (Experiment 3). The line on each boxplot denotes the median score (also presented numerically), the edges of the box denote the 25th and 75th percentile, and the whiskers extend the data points that lie within 1.5 times the interquartile range (IQR). The black dots represent outliers based on the interquartile range



when compared to Grade 1 or 2 children, $W=153$, $p=0.26$ and $W=223$, $p=0.20$, respectively.¹⁵

General discussion

Memory abilities are typically viewed as fundamental for both general communication abilities such as language comprehension and production and scholastic abilities such as reading, mathematics, and problem solving (e.g., Arrington et al. 2014; Gathercole et al. 2004, 2006; Swanson and Jerman 2007). Without contradicting this relationship, the present results suggest that the converse relationship also holds true, namely, that growth of memory abilities, at least of verbal order memory, depends on exposure to formal schooling, and particularly on literacy acquisition.

The results of the first experiment ran on children suggest a link between literacy acquisition and serial-order memory, as correlations between performance on serial-order reconstruction and literacy scores remained significant after controlling for the effect of age. In both first- and second-graders, there were also significant within-group correlations between literacy scores and performance on serial-order

reconstruction, although age only slightly varied within each one of these groups.

Experiment 2 further showed that, at the end of the school year, six-year-old first graders (Y1) significantly outperformed age-matched kindergarteners (OK) on serial-order reconstruction. In addition, in first grade, the children's age (younger vs. older: Y1 vs. O1) did not affect memory performance. There was thus a schooling effect without age effect. Distinguishing the precise source(s) of this schooling effect is far from being obvious, as multiple factors in the school environment are susceptible to play a role. As discussed by Ardila et al. (2010), attending school contributes to the development of specific attitudes toward knowing, understanding, and thinking. School favors indeed specific principles such as learning, memorizing, understanding, and achieving, all having some important consequences on cognition and, in particular, on memory skills. Another source of memory enhancement might be reading acquisition, which takes generally place at the beginning of formal schooling (Demoulin and Kolinsky 2016). In line with this idea, we observed that serial-order reconstruction performance was strongly related to children's literacy abilities, even after controlling for the effects of non-verbal reasoning and receptive vocabulary.

The third experiment, ran on adults of varying (correlated) levels of schooling and literacy, confirmed the idea that literacy acquisition does benefit serial-order memory. Indeed, adults who have had

¹⁵ In the comparison with kindergarteners and Grade 1 children, the average literacy score did not include the reading scores, as the reading tests were only presented to a minority of these children (see Table 1).

less or no opportunity in their life to go to school (–schooled), and hence who remained poorly literate or illiterate, presented poorer serial-order reconstruction performance than those who attended school for more than 12 years and hence were more fluent readers (+schooled). The idea that literacy plays an important role in this group difference was supported by a significant correlation between serial-order reconstruction performance and the literacy score, even when accounting for the age difference between groups. In addition, it is particularly striking that the –schooled adults’ order reconstruction performance was closer to the performance of the children examined in Experiment 1 than to the one of the +schooled adult participants. Whereas the –schooled adults presented slightly better literacy and serial-order reconstruction performances than the kindergartners, they differed from Grade 1 and 2 children neither on serial-order reconstruction nor on literacy performance. This most probably corresponds to the fact that the –schooled adults had attended school (presumably of poor pedagogical quality) for only about 2 years on the average. Thus, although some aspects of memory certainly improve with maturation (see discussion in e.g., Cowan 2016), in the present case, when both the task and the material are easy for all age groups, it seems that it is the power of the schooling experience, including literacy acquisition, that shapes memory for serial order.

This conclusion is in agreement with the claims made by Smalle et al. (2019). These researchers reported Devanagari (schooled) literates to present better serial-order reconstruction performance than illiterate adults and suggested that the improved capacity of the literate group is a consequence of literacy acquisition. The present study further suggests that Smalle et al.’s results were not due to the peculiarities of the Devanagari script. Rather, taken together, both studies provide further empirical support for the role of schooling and its related experiences, such as literacy acquisition, on the development of verbal memory.

Our results are also consistent with former studies showing an impact of literacy acquisition on other aspects of verbal STM and working memory (e.g., Castro-Caldas et al. 1998; Kosmidis et al. 2011; Matute et al. 2012; Morais et al. 1986; Nation and Hulme 2011; Silva et al. 2012; for discussion, see Demoulin and Kolinsky 2016). As a matter of fact,

the results of Experiment 3 clearly showed that the effect of schooling and/or literacy is not limited to memory for serial order: performance on a delayed single nonword repetition task that was aimed at maximizing processing requirements for item information while minimizing serial-order processing requirements also differed between the –schooled and +schooled participants, and was strongly correlated to literacy level.

Further work should thus be aimed at examining exactly *how* schooling and/or reading acquisition enhances performance in verbal memory tasks. As discussed by Demoulin and Kolinsky (2016), one possibility is that it improves the processing and the temporary encoding of spoken language through phoneme awareness. Yet, in the present study, in both children and adults, the correlation between reading level and serial-order STM remained significant after controlling for the effect of phoneme awareness. Furthermore, in adults, even the correlation between reading and item STM remained significant after controlling for the effect of phoneme awareness. Thus, whereas phoneme awareness seems related to verbal STM performance, other factors are at play. In addition, contrary to what had been reported by Binamé and Poncelet (2016), in the present study we observed significant correlations between serial-order memory and both word and pseudoword reading fluency.

In any case, the present study offers an account for the developmental memory shift reported in the literature as occurring around 6–7 years of age. Indeed, verbal memory performance increases steeply from the preschool years up to 8 years of age, and more gradually afterward (e.g., Gathercole 1999). The present results suggest that it is not maturation per se but experiences at school, and in particular literacy acquisition, that explain this performance jump.

The importance of school experience does not exclude the impact of differences in socioeconomic background. As a matter of fact, in Experiment 1, serial-order reconstruction was better in children from SES+ schools compared to children from SES– schools, in particular before formal learning at school had begun, and there were significant correlations between serial-order reconstruction performance and parental education level. Thus, contrary to what has been claimed with respect to several measures of

STM and working memory (Engel et al. 2008), serial-order memory tests are not impervious to differences in socioeconomic background. In fact, what our data suggest is that early school contributes to attenuate the effects of a low SES, as the only significant SES-related difference on serial-order reconstruction was observed in kindergarteners. Yet, more evidence is necessary on this point, as that, despite their compensatory role, schools are not the ultimate equalizer and the socioeconomic gradient may persist despite educational attainment (e.g., Brownell et al. 2006).

The present data thus argue for a role of both schooling and/or literacy and other experiences linked to socioeconomic background in shaping verbal serial-order memory and, more generally, verbal STM. This conclusion ought however be contextualized in at least two ways.

First, the effects observed in the present study probably reflect specific impacts of Western-style schooling and/or literacy education. Indeed, as discussed by Demoulin and Kolinsky (2016), the impact of literacy and/or schooling depends on qualitative aspects of the instruction. For instance, in the study run by Scribner and Cole (1981) in Liberia, adults who were literate in Arabic because they had attended Quranic schools in their youth did not present a general memory advantage in comparison to Vai (tribal) literates who had acquired their indigenous (syllabic) script at home through individual tuition; for example, they were not better on free recall of words. Yet, they presented a specific advantage for ordered recall of word lists. This probably reflects the influence of the incremental method of learning the Qur'an used in the Islamic schools of that region, in which tuition included a large amount of memorization and group recitation of verses, typically by repeating sentences lengthened one word at a time at each repetition. A study on children attending Quranic preschool in Morocco, in which the curriculum also includes a large amount of memorization and group recitation of verses, supports this idea: it showed a positive effect of this type of schooling on serial memory (digit span, name span, and incremental versions of these tests) that did not generalize to other kinds of memory skills, such as discourse memory and ordered memory of pictures (Wagner and Spratt 1987).

Second, our conclusion holds true only as regards literate societies. As discussed for instance by Ong (1982, p. 34), primary oral cultures use quite specific strategies of preserving information in the absence of writing: “Your thought must come into being in heavily rhythmic, balanced patterns, in repetitions or antitheses, in alliterations and assonances, in epithetic and other formulaic expressions, in standard thematic settings (the assembly, the meal, the duel, the hero's ‘helper’, and so on), in proverbs which are constantly heard by everyone so that they come to mind readily and which themselves are patterned for retention and ready recall, or in other mnemonic form.” In other words, “In a primary oral culture, to solve effectively the problem of retaining and retrieving carefully articulated thought, you have to do your thinking in mnemonic patterns, shaped for ready oral recurrence” (Ong, *ibid.*). In Westernized societies, however, these characteristics are not exploited anymore, with the exception of some traditional activities like oral poetry (e.g., Curran 2014). In addition to examining memory performance in the few primary nonliterate societies still existing, studying the memory capacities of illiterate poets (in addition to examining their metaphonological abilities, as did Cary 1990; see also Morais 1994) would be a way of evaluating the impact of other activities on verbal memory.

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Compliance with ethical standards

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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